From Classroom/Integrated School Programs 7 in Top 20, 23 in Top 100, 54 in Top 300, 106 in Top 500 All India Ranks & 2314 Students from Classroom /Integrated School Programs & 3723 Students from All Programs have been Awarded a Rank in JEE (Advanced), 2013

FIITJEE

JEE(Advanced), 2014

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS

FULL TEST –IV
(Paper-2)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	C	C	C
2.	C	D	B
3.	B	A	A
4.	C	B	D
5.	A, B, D	A, B, D	A, B, C
6.	B, C	A, B, C	B, C
7.	A, B, C, D	A, B	B, C
8.	B, C	A, C	A, C
9.	A, B, C, D	A, B, C	B, C
1.	(A)→ (q, r) (B)→ (s) (C)→ (p) (D)→ (t)	A → (q, r) B → (p) C → (t) D → (r, s)	(A) → (t), (B) → (p), (C) → (s), (D) → (q)
2.	(A) → (p, q) (B) → (q, r, s) (C) → (p, q, r) (D) → (q, r, s)	A → (q, r) B → (q, t) C → (p, r) D → (p, s)	(A) → (p, r), (B) → (p, r), (C) → (q), (D) → (q, r, s, t)
1.	5	5	2
2.	3	1	4
3.	3	2	6
4.	4	3	2
5.	8	2	4
6.	3	2	6
7.	1	7	0
8.	5	4	0

Physics

PART – I

2. $f_{max} = f_{1_{max}} + f_{2_{max}} = 150 + 450 = 600N$

Hence A and B have no acceleration but block C has a tendency to move down. Hence T_1 = 500 N, f_2 = 450 N, T_2 = 50 N, T_3 = 50 N, f_1 = 0 N

4. $\omega = \frac{v_0}{\ell}$ and $v_{cm} = \frac{3}{2}v_0$

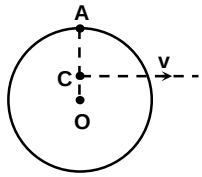
$\Rightarrow L = M\left(\frac{\ell}{2}\right)\times\frac{3}{2}v_0 + \frac{M\ell^2}{12}\times\frac{v_0}{\ell}$

$= mv_0\ell\left[\frac{3}{4}+\frac{1}{12}\right] = \frac{5}{6}mv_0\ell$

5. $K = K_D + K_P$

$= \frac{1}{2}mv_0^2\left(1+\frac{1}{2}\right)+\frac{1}{2}m(2V_0)^2$

$= \frac{11}{4}mV_0^2$



7. Use rules of combination of resistances and distribution of currents

8. $E_0z^2\left(1-\frac{1}{9}\right)-E_0z^2\left(\frac{1}{4}-\frac{1}{9}\right)=3E_0$

$z = 2$

$\lambda_1/\lambda_2 = 3$

$KE_1 = E_0\left(1-\frac{1}{9}\right)-\phi$

$KE_2 = E_0z^2\left(1-\frac{1}{4}\right)-\phi$

$KE \propto \frac{1}{\lambda^2}$ = 8.5 eV.

SECTION - B

1. Velocity of longitudinal wave = $\sqrt{\frac{2\times10^{12}}{5\times10^3}} = 20000$ m/s

In situation (p) $\frac{5\lambda}{4}=1m$ $\therefore$ $\lambda = \frac{4}{5}m$ $\therefore$ f = 25 kH_2

In situation (q) $\frac{\lambda}{2}=1m$ $\therefore$ λ = 2m $\therefore$ f = 10 kH_2

In situation (r) $\frac{\lambda}{2}=1m$ $\therefore$ λ = 2m $\therefore$ f = 10 kH_2

In situation (s) $\frac{\lambda}{4}=\frac{1}{3}m$ $\therefore$ $\lambda = \frac{4}{3}m$ $\therefore$ f = 15 kH_2

In situation (t) $\frac{\lambda}{4}=\frac{1}{3}m$ $\therefore$ $\lambda = \frac{4}{9}m$ $\therefore$ f = 45 kH_2

SECTION –C

1. $\Delta x = (S_1P - t) + \mu_1 t - S_2P$
$0 = (S_1P - S_2P) + (\mu_1 - 1)t$
$\Rightarrow$ $d \sin\theta = (\mu_1 - 1)t$

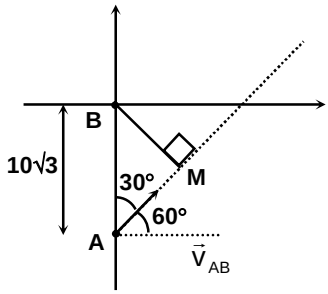
$$\frac{y_1 d}{D} = (\mu_1 - 1)t$$

$$\frac{y_2 d}{D} = (\mu_1 - 1)t$$

$$y_1 - y_2 = \frac{(\mu_1 - \mu_2)tD}{d}$$

$= 5mm$

2. As observed by B motion of A is along AM and BM is the shortest distance between them. Relative displacement of A w.r.t to B is AM = AB cos 30°



$\vec{V}_{AB} = 5\hat{i} + 5\sqrt{3}\hat{j}$

Time taken (t) = $\frac{|\vec{S}_{AB}|}{|\vec{v}_{AB}|} = \frac{AM}{v_{AB}} = \frac{10\sqrt{3}\cos 30°}{10}$

$= 1.5$ sec

3. $L_1 = \frac{\mu_0 N_1^2 A}{\ell}$, $L_2 = \frac{\mu_0 N_2^2 A \mu_r}{\ell}$

Since $V = -L\frac{dI}{dt}$, So

$$\frac{V_1}{V_2} = \frac{L_1}{L_2} = \left(\frac{N_1}{N_2}\right)^2 \frac{1}{\mu_r} = \left(\frac{600}{200}\right)^2 \times \frac{1}{3} = 3$$

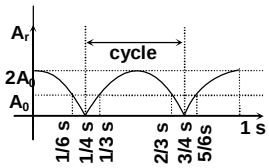
4. The half of the maximum current is equal to

$$i_1 = \frac{B\omega\ell^2}{4R} \text{ at } t = \frac{L}{R}\ln 2$$

The torque at this instant = $\frac{B^2\omega\ell^4}{8R}$ = 4 N-m

5. $y_1 = A \sin \omega_1 t$
$y_2 = A \sin \omega_2 t$

$$y_r = 2A\cos\left\{\frac{(\omega_2 - \omega_1)}{2}t\right\}\left\{\sin\frac{(\omega_2 + \omega_1)}{2}t\right\}$$



Resultant amplitude $A_r = 2A_0|\cos(\Delta\omega)t/2|$

$$(\Delta\omega)\frac{t}{2} = \frac{\pi}{2} \Rightarrow t = \frac{1}{4}s$$

$$(\Delta\omega)\frac{t}{2} = \frac{\pi}{3} \Rightarrow t = \frac{1}{6}s$$

In one cycle of intensity of 1/2s, the detector remain idle for

$2\left(\frac{1}{4}-\frac{1}{6}\right)s=\frac{1}{6}\sec$

$\therefore$ In ½ sec cycle, active time is $\left(\frac{1}{2}-\frac{1}{6}\right)$ = 1/3 sec

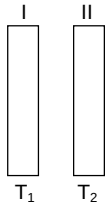
$\therefore$ In 12 sec interval, active time is $12\times\frac{(1/3)}{(1/2)}$ = 8 sec

6. $P = 2\sigma AT_1^4 - \sigma AT_2^4$ (steady state condition of sheet I) …(i)

and $2\sigma AT_2^4 = \sigma AT_1^4$ (steady state condition of sheet II) …(ii)

$\Rightarrow$ $P = 3\sigma AT_2^4$

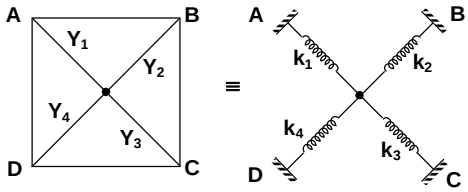
= 3 kW



7. $T = \pi\left[\sqrt{\frac{m\ell}{9YA}}+\sqrt{\frac{m\ell}{16YA}}\right]$

$= \pi\left[\sqrt{\frac{m\ell}{YA}}\right]\left[\frac{1}{3}+\frac{1}{4}\right]$

$= \frac{7\pi}{12}\sqrt{\frac{m\ell}{YA}}$



$m = 144\times10^{-3}$ kg

ℓ = AO = BO = CO = DO = 10×10^{-2} m

$A = \pi^2\times10^{-6}$ m^2

$Y = 4.9\times10^9$

$T = \frac{7\pi}{12}\sqrt{\frac{144\times10^{-3}\times10\times10^{-2}}{\pi^2\times10^{-6}\times49\times10^8}} = \frac{7\pi}{12}\times\frac{12}{\pi}\times\frac{1}{7}\sqrt{\frac{10^{-2}}{10^8}}$

$= 1\times10^{-3}$

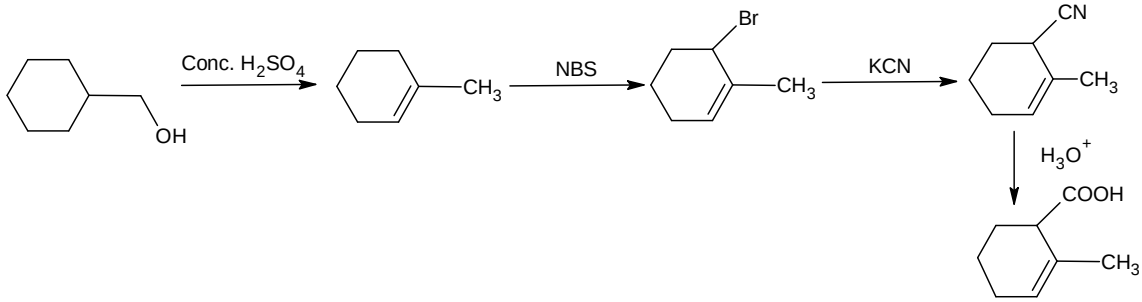
= 1 ms

Chemistry

PART – II

SECTION – A

1.



2.

	$2AB_3(g)$	$\rightleftharpoons$	$2AB_2(g)$	+	B_2
t = 0	0.1 mol/lt		0		0
t = eq	0.1 – 0.08		0.08		0.04

Total no. of moles per litre (n/v) = 0.14

$$P = \frac{n}{V} \times R \times T = 0.14 \times 0.082 \times 546 = 6.25 \text{ atm}$$

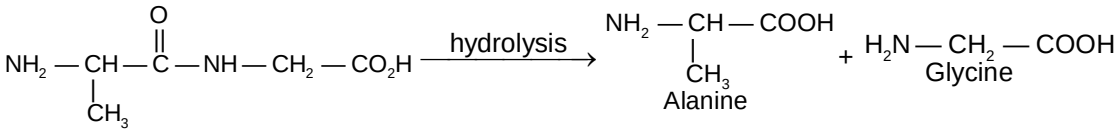
3. In Al_4C_3, three moles of CH_4 are formed.

4. PF_5 has sp^3d hybridization (trigonal bipyramid)
BrF_5 has sp^3d^2 hybridization (square pyramidal).

5. In case of (C) racemic mixture is obtained.

7. Fact → Chromyl chloride test is not given by chlorides of Hg, Sn, Ag, Pb and Sb.

8.



9.

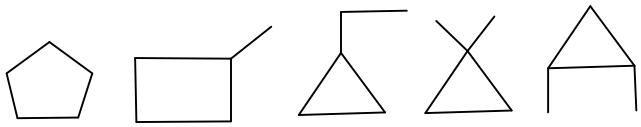
$$2NH_4ClO_4 \longrightarrow N_2 + Cl_2 + 2O_2 + 4H_2O$$

$$IO_3^- + N_2H_4 + 2H^+ \longrightarrow I^+ + N_2 + 3H_2O$$

$$HN_3 + HNO_2 \longrightarrow N_2 + N_2O + H_2O$$

$$2NH_2OH + 4CuO \longrightarrow N_2O + 2Cu_2O + 3H_2O$$

SECTION – C

1. 

2. $\text{E.W.} = \dfrac{\text{M.W}}{\text{acidity}}$

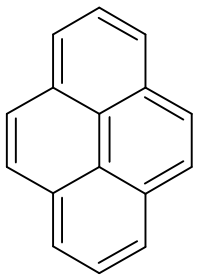
   $Zn(OH)_2$ can replace only one OH.

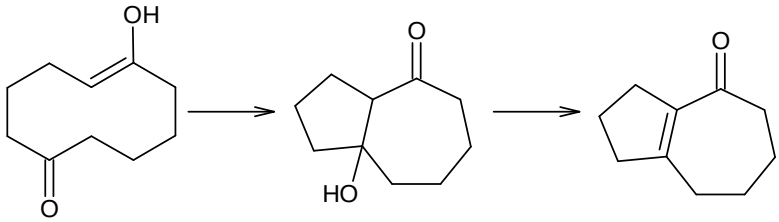
3. $[Cu(NH_3)_4]SO_4 \rightleftharpoons [Cu(NH_3)_4]^{2+} + SO_4^{2-}$

4. $\dfrac{(n_2 - n_1)(n_2 - n_1 + 1)}{2}$

5. $\text{rate} \propto \dfrac{1}{[A]}$

6. Pinacol-pinacolone reaction

7.  The central pi-bond is not in conjugation.

8. 

Mathematics

PART – III

SECTION – A

1. We have $b_k = \frac{n}{2}(a_K + a_{n+k-1}) = \frac{n}{2}[a_1 + (k-1)d + a_1 + (n+k-2)d]$

$= \frac{n}{2}[a_n + a_1 + 2(k-1)d]$ (d is the common difference)

$\Rightarrow \sum_{k=1}^{n} b_k = \frac{n}{2}\left[na_n + na_1 + 2d\sum_{k=1}^{n}(k-1) \right] = \frac{n}{2}[na_n + na_1 + d(n-1)n]$

$= \frac{n^2}{2}[a_n + a_1 + (n-1)d] = \frac{n^2}{2}[a_n + a_n] = n^2 a_n$

2. Here $z - 1 = e^{i\theta}$

$\Rightarrow z = 2\cos^2\frac{\theta}{2} + 2i\sin\frac{\theta}{2}\cos\frac{\theta}{2} = 2\cos\frac{\theta}{2}e^{i\theta/2}$

and $z - 2 = z - 1 - 1 = e^{i\theta} - 1 = -1 + e^{i\theta} = -2\sin^2\frac{\theta}{2} + 2i\sin\frac{\theta}{2}\cos\frac{\theta}{2} = 2i\sin\frac{\theta}{2}e^{i\theta/2}$.

Hence $2i\sin\frac{\theta}{2}e^{i\theta/2} = k \cdot 2\cos\frac{\theta}{2}e^{i\theta/2}\tan\frac{\theta}{2}$

$\Rightarrow k = i$.

3. $\angle OAB = \angle APO = \frac{\theta}{2}$

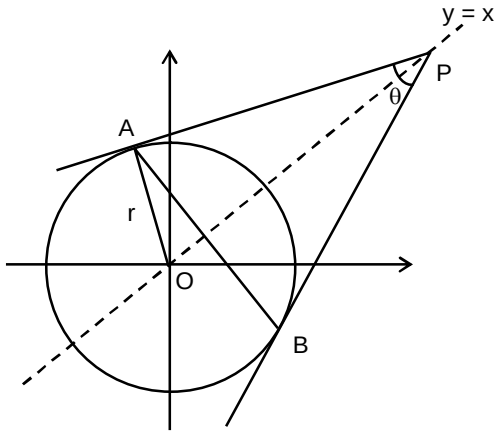
$\therefore$ AP = BP = r $\cot\frac{\theta}{2}$ and

AB = 2r $\cos\frac{\theta}{2}$.

$\therefore$ Perimeter of ΔAPB, $L = 2r\left(\cot\frac{\theta}{2} + \cos\frac{\theta}{2}\right)$

$\frac{dL}{dr} = r\left(-\operatorname{cosec}^2\frac{\theta}{2} - \sin\frac{\theta}{2}\right) < 0$

for all $\theta \in \left[\frac{\pi}{3}, \frac{2\pi}{3}\right]$.



$\therefore$ L is maximum when $\theta = \frac{\pi}{3}$ and then OP = r cosec $\frac{\theta}{2}$ = 2r.

So, the co-ordinates of P are (2r cos 45°, 2r sin 45°).

4. The line is $z(1 + i) = \bar{z}(1 - i)$ or $z(-1 + i) + \bar{z}(-1 - i) = 0$.

If the reflection of a number ω in the line $a\bar{z} + \bar{a}z + b = 0$, $b \in R$ be ω', then

$\omega'\bar{a} + \bar{\omega}a + b = 0$

Now $\omega = \frac{2+i}{3-i} = \frac{(2+i)(3+i)}{10} = \frac{1+i}{2}$

So, $\omega'(-1+i) + \left(\frac{1-i}{2}\right)(-1-i) = 0$

$\therefore\ \omega'(-1+i) = \dfrac{(1-i)(1-i)}{2} = 1$

$\Rightarrow\ \omega' = \dfrac{1}{-1+i} = \dfrac{-1-i}{2} = \dfrac{2+i}{-3+i}$

5. x = P(A wins) = P(H) + P(TH) + P(TTTH) + P(TTTTH) + …..
= p + qp + q^3p + q^4p + ….. (q = 1 – p)
= (p + q^3p + q^6p + …..) + (qp + q^4p + q^7p + …..)

= $\dfrac{p}{1-q^3} + \dfrac{pq}{1-q^3} = \dfrac{1+q}{1+q+q^2}$ and y = 1 – x = $\dfrac{q^2}{1+q+q^2}$

Now 0 < p < 1 $\Rightarrow$ 0 < q < 1, so, 1 + q > q^2 $\Rightarrow$ x > y

Also, x > y $\Rightarrow$ 2x > x + y = 1 or x > $\dfrac{1}{2}$ thus y < $\dfrac{1}{2}$

6. In ΔABC, b + c – a > 0, c + a – b > 0, a + b – c > 0, so

$$\frac{(b+c-a)+(c+a-b)+(a+b-c)}{3} \geq \{(b+c-a)(c+a-b)(a+b-c)\}^{1/3}$$

$\Rightarrow P^3 \geq 27(b + c – a)(c + a – b)(a + b – c)$

Also,

$$\frac{(a+b+c)+(b+c-a)+(c+a-b)+(a+b-c)}{4} > \{(a+b+c)(b+c-a)(c+a-b)(a+b-c)\}^{1/4}$$

$\Rightarrow \dfrac{2P}{4} > (16A)^{1/4}$ $\Rightarrow P > 4A^{1/4}$ or $P^4 > 256A$

Next, we know for a given parameter, equilateral triangle has the largest area, so the area of triangle

$A \leq \dfrac{\sqrt{3}}{4}\left(\dfrac{p}{3}\right)^2$ (for equilateral triangle, a = b = c = p/3)

Now, $p^2 = (a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$
$= 3(a^2 + b^2 + c^2) – 2(a^2 + b^2 + c^2 – ab – bc – ca)$
$\leq 3(a^2 + b^2 + c^2)$ (equality holds iff a = b = c)

Thus $A \leq \dfrac{\sqrt{3}}{4}\dfrac{a^2+b^2+c^2}{3}$ $\Rightarrow a^2 + b^2 + c^2 \geq 4\sqrt{3}\,A$

7. Let the equation of the required plane be $\dfrac{x}{a} + \dfrac{y}{b} + \dfrac{z}{c} = 1$

The points A, B, C are (a, 0, 0), (0, b, 0) and (0, 0, c) and the centroid of ΔABC is $\left(\dfrac{a}{3}, \dfrac{b}{3}, \dfrac{c}{3}\right)$

Thus a = 3, b = 3r, $c = 3r^2$, $r \neq 0$

So, the equation of plane is $\dfrac{x}{3} + \dfrac{y}{3r} + \dfrac{z}{3r^2} = 1$

It passes through (t, t, t)
so, $(t – 3)r^2 + rt + t = 0$, clearly $t \neq 0$
for real r, $t^2 – 4(t – 3)t \geq 0$, $t \neq 3$ $\Rightarrow 0 < t \leq 4$, $t \neq 3$
when t = 3, r = – 1, so $0 < t \leq 4$

8. We have $\sin(\cot^{-1}\cos\tan^{-1}x) = \sin(\cot^{-1}\cos\theta)$, where $\theta = \tan^{-1}x$

= $\sin\cot^{-1}\dfrac{1}{\sqrt{1+\tan^2\theta}} = \sin\cot^{-1}\dfrac{1}{\sqrt{1+x^2}} = \sqrt{\dfrac{1+x^2}{2+x^2}}$

Now for all $x \in R$, $0 < \frac{1+x^2}{2+x^2} < 1$, then $[\sin(\cot^{-1} \cos \tan^{-1} x)] = \left[\frac{1+x^2}{2+x^2}\right] = 0$

$\therefore$ The equation is $\sin\theta = 0 \Rightarrow \theta = n\pi$, $n \in I$

9. We have $a\cos x - (ax + b)\sin x + c\sin x + (cx + d)\cos x = x\cos x \ \forall\ x \in R$
$\Rightarrow (a + d)\cos x + (c - b)\sin x - ax\sin x + (c - 1)x\cos x = 0$ is an identity
$\Rightarrow a + d = 0, c - b = 0, a = 0, c - 1 = 0 \Rightarrow a = d = 0$ and $b = c = 1$

SECTION – B

1. **(A)** Let the sides a and b are the roots of $x^2 - 5x + 3 = 0$, then $a + b = 5$ and $ab = 3$

Also, $\angle C = \frac{\pi}{3}$

$\therefore \frac{1}{2} = \frac{a^2 + b^2 - c^2}{2ab} \Rightarrow ab = (a + b)^2 - 2ab - c^2 \Rightarrow c = 4$

Now, $rR = \frac{abc}{2(a+b+c)} = \frac{3\times 4}{2(5+4)} = \frac{2}{3}$

(B) We have $a = \frac{r_1(r_2 + r_3)}{\sqrt{r_1r_2 + r_2r_3 + r_3r_1}} = \frac{8\times 36}{24} = 12$

(C) Given $1 \le a \le 10$
Also, for roots of opposite sign $2a^2 - 7a + 3 < 0$
or $\frac{1}{2} < a < 3 \Rightarrow a = 1$ or 2

$\therefore$ Desired probability $= \frac{2}{10} = \frac{1}{5}$

(D) (α, β) lies on the director circle of the ellipse i.e. on $x^2 + y^2 = 9$
So, we can assume $\alpha = 3\cos\theta$, $\beta = 3\sin\theta$.
Thus $F = 12\cos\theta + 9\sin\theta = 3(4\cos\theta + 3\sin\theta)$
$\Rightarrow -15 \le F \le 15$

2. **(A)** Let consecutive coefficients be ${}^nC_r$ and ${}^nC_{r+1}$. Then, $\frac{n!}{(n-r)!r!} = \frac{n!}{(n-r-1)!(r+1)!}$

$\Rightarrow \frac{1}{(n-r)(n-r-1)!r!} = \frac{1}{(n-r-1)!(r+1)r!}$

$\Rightarrow r + 1 = n - r$

$\Rightarrow n = 2r + 1$ hence, n is odd.

(B) $E = (19 - 4)^n + (19 + 4)^n$
$2[{}^nC_0 19^n + {}^nC_2 19^{n-2}4^2 + \ldots.. + {}^nC_n 4^n]$ when n is even or
$2[{}^nC_0 19^n + {}^nC_2 \cdot 19^{n-2}\cdot 4^2 + \ldots.. + {}^nC_{n-1} 19\cdot 4^{n-1}]$ when n is odd
$\Rightarrow$ E is divisible by 19 when n is odd

(C) ${}^{10}C_0\ {}^{20}C_{10} - {}^{10}C_1\ {}^{18}C_{10} + {}^{10}C_2\ {}^{16}C_{10} - \ldots..$
= Coefficient of x^{10} in $[{}^{10}C_0(1 + x)^{20} - {}^{20}C_1 \times (1 + x)^{18} + {}^{10}C_2(1 + x)^{16} - \ldots..]$
= Coefficient of x^{10} in $[{}^{10}C_0((1 + x)^2)^{10} - {}^{10}C_1 \times ((1 + x)^2)^9 + {}^{10}C_2((1 + x)^2)^8 - \ldots..]$
= Coefficient of x^{10} in $[(1 + x)^2 - 1]^{10}$
= Coefficient of x^{10} in $[2x + x^2]^{10}$
$= 2^{10}$

(D) $T_r = {}^{14}C_{r-1}x^{r-1}$; $T_{r+1} = {}^{14}C_r x^r$; $T_{r+2} = {}^{14}C_{r+1}x^{r+1}$
By the given condition, $2\ {}^{14}C_r = {}^{14}C_{r-1} + {}^{14}C_{r+1}$

$$\Rightarrow 2 = \frac{^{14}C_{r-1}}{^{14}C_r} + \frac{^{14}C_{r+1}}{^{14}C_r}$$

$$\Rightarrow 2 = \frac{r}{14-r+1} + \frac{14-(r+1)+1}{r+1}$$

$$\Rightarrow 2 = \frac{r}{15-r} + \frac{14-r}{r+1}$$

$\Rightarrow r = 9$

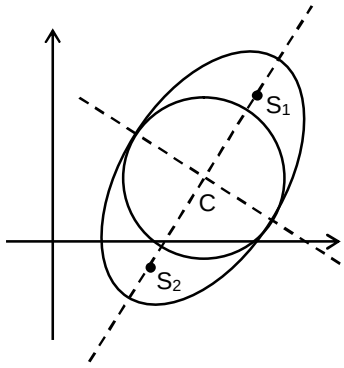
SECTION – C

1. $f'(x) = 6x^2 - 18ax + 12a^2 = 0 \Rightarrow x = a, 2a.$
Given $x_2 = x_1^2$, so a cannot be negative. If a > 0, then local minimum occurs at x = 2a and local maximum at x = a. Thus $x_1 = a$ and $x_2 = 2a$
$\Rightarrow 2a = a^2 \Rightarrow a = 2$ (if a = 0, f(x) is always increasing)

2. Since x, y, z are in A.P., x + z = 2y and hence
$$\frac{x+y}{2y-x} + \frac{y+z}{2y-z} = \frac{x+y}{z} + \frac{y+z}{x} = \frac{y}{x} + \frac{y}{z} + \frac{x}{z} + \frac{z}{x}$$
$$= \frac{x}{z} + \frac{x+z}{2z} + \frac{x+z}{2x} + \frac{z}{x} = 1 + \frac{3}{2}\left(\frac{x}{z} + \frac{z}{x}\right)$$
$\geq 1 + 3$ [A. M. $\geq$ G.M.]
$\Rightarrow$ minimum value of the given expression is 4.

3. $24300 = 2^2\, 3^5\, 5^2$
Number of odd divisors = the number of terms in $(1 + 3 + 3^2 + 3^3 + 3^4 + 3^5)(1 + 5 + 5^2)$
$= 6 + 6 + 6 = 18$
This also includes the factor 1. So the number of proper odd divisors = 17

4. $|z - 3 - i| = 2$ represents a circle with centre 3 + i and radius 2
$|z - 2 + i| + |z - 4 - 3i| = 6$ represents an ellipse with centre $\left(\frac{2-i+4+3i}{2}\right) = 3 + i$ and major axis = 6
Also, the distance between foci = $2\sqrt{5}$
$\Rightarrow$ eccentricity = $\frac{2\sqrt{5}}{6} = \frac{\sqrt{5}}{3} < 1$
$\therefore$ Minor axis = $2 \times 3\sqrt{1 - \frac{5}{a}} = 4$
$\because$ diameter of circle = minor axis
So, the circle touches the ellipse internally



5. Normal to hyperbola at t is $y - \frac{c}{t} = t^2(x - ct)$ ..... (1)
It touches $xy = -c^2$
Solving with (1), we get $x\left\{\frac{c}{t} + t^2(x - ct)\right\} + c^2 = 0 \Rightarrow t^2x^2 + \left(\frac{c}{t} - ct^3\right)x + c^2 = 0$
For equal roots, we have $\left(\frac{1}{t} - t^3\right)^2 - 4t^2 = 0 \Rightarrow (1 - t^4)^2 - 4t^4 = 0$
$\Rightarrow t^4 - 1 - 2t^2 = 0$ or $t^4 - 1 + 2t^2 = 0$

$\Rightarrow t^2 = \frac{2 \pm \sqrt{8}}{2}$ or $\frac{-2 \pm \sqrt{8}}{2} \Rightarrow t^2 = 1 + \sqrt{2}$ or $-1 + \sqrt{2}$

6. $|z - 3| \le |z - 1| \Rightarrow z + \bar{z} \ge 4 \Rightarrow x \ge 2$ $(\because z + \bar{z} = 2x)$
$|z - 3| \le |z - 5| \Rightarrow z + \bar{z} \le 8 \Rightarrow x \le 4$
$|z - i| \le |z + i| \Rightarrow i(z - \bar{z}) \le 0 \Rightarrow y \ge 0$ $(\because z - \bar{z} = 2iy)$
$|z - i| \le |z - 5i| \Rightarrow i(z - \bar{z}) \ge -6 \Rightarrow y \le 3.$
Clearly the region is a rectangle of area $= 2 \times 3 = 6$

7. The given function is

$$y = \cot^{-1}\left(\frac{1+2\ln x}{1-2\ln x}\right) + \tan^{-1}\left(\frac{3+2\ln x}{1-6\ln x}\right).$$

Let $\tan\theta = 2\ln x$, $\tan\phi = 3$ so that

$$y = \cot^{-1}\left(\frac{\tan\frac{\pi}{4} + \tan\theta}{1 - \tan\frac{\pi}{4}\tan\theta}\right) + \tan^{-1}\left(\frac{\tan\phi + \tan\theta}{1 - \tan\phi\tan\theta}\right)$$

$$= \cot^{-1}\left(\tan\left(\frac{\pi}{4} + \theta\right)\right) + \tan^{-1}\left(\tan\left(\phi + \theta\right)\right)$$

$$= \frac{\pi}{2} - \left(\frac{\pi}{4} + \theta\right) + \phi + \theta = \frac{\pi}{4} + \phi = \frac{\pi}{4} + \tan^{-1}3 \text{ = constant}$$

$\Rightarrow \quad \frac{d^2y}{dx^2} = 0$.

8. We have $f(x) + f(x + 4) = f(x + 2) + f(x + 6)$
$\Rightarrow f(x + 2) + f(x + 6) = f(x + 4) + f(x + 8)$
$\Rightarrow f(x) = f(x + 8)$, and hence $g'(x) = f(x + 8) - f(x) = 0$